Faraday Rotation

$$\vec{E}(z,t) = E_{0x}\, e^{i(kz-\omega t)}\,\hat{x} + E_{0y}\, e^{i(kz-\omega t)}\,\hat{y}$$

$$\ddot{\vec{r}} = -\omega_0^2\,\vec{r} + \frac{q}{m}\,\dot{\vec{r}}\times\vec{B} + \frac{q}{m}\,\vec{E}$$

Electrons m external static field $\vec{B}$ (this is not the field of the EM wave which we neglect in non-relativistic limit)

$$\begin{cases} \ddot{x} + \omega_0^2 x - \frac{qB}{m}\dot{y} = \frac{q}{m} E_{0x}\, e^{-i\omega t} \\ \ddot{y} + \omega_0^2 y + \frac{qB}{m}\dot{x} = \frac{q}{m} E_{0y}\, e^{-i\omega t} \end{cases}$$

$$\vec{r}\times\vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ x & y & 0 \\ 0 & 0 & B \end{vmatrix} = \hat{x}\, yB - \hat{y}\, xB$$

$$x = x_0\, e^{-i\omega t}, \quad y = y_0\, e^{-i\omega t}$$

Assume, as usual, large wavelength $\Rightarrow$ can neglect z-dependence of $\vec{E}$.

$$(\omega_0^2-\omega^2)\, x_0 + i\omega\Omega_c\, y_0 = \frac{qE_{0x}}{m}$$

$$(\omega_0^2-\omega^2)\, y_0 - i\omega\Omega_c\, x_0 = \frac{qE_{0y}}{m}$$

$\Omega_c = \frac{qB}{m}$ cyclotron frequency

<u>Case of conductor</u>: $\omega_0 = 0$

$$\begin{pmatrix} -\omega^2 & i\omega\Omega_c \\ -i\omega\Omega_c & -\omega^2 \end{pmatrix}\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \frac{q}{m}\begin{pmatrix} E_{0x} \\ E_{0y} \end{pmatrix}$$

$$\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} \frac{-1}{\omega^2-\Omega_c^2} & \frac{-i\Omega_c}{\omega(\omega^2-\Omega_c^2)} \\ \frac{i\Omega_c}{\omega(\omega^2-\Omega_c^2)} & \frac{-1}{\omega^2-\Omega_c^2} \end{pmatrix}\left(\frac{q}{m}\right)\begin{pmatrix} E_{0x} \\ E_{0y} \end{pmatrix}$$

For $\omega > \omega_p \gg \Omega_c$ (consider propagating wave)

$$\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = -\frac{q}{m\omega^2}\begin{pmatrix} 1 & -\frac{i\Omega_c}{\omega} \\ \frac{i\Omega_c}{\omega} & +1 \end{pmatrix}\begin{pmatrix} E_{0x} \\ E_{0y} \end{pmatrix}$$

$$\Rightarrow \quad x(t) = -\frac{q}{m\omega^2}\left(E_{0x}\, e^{-i\omega t} - \frac{i\Omega_c}{\omega} E_{0y}\, e^{-i\omega t}\right)$$

$$y(t) = -\frac{q}{m\omega^2}\left(E_{0y}\, e^{-i\omega t} + \frac{i\Omega_c}{\omega} E_{0x}\, e^{-i\omega t}\right)$$

$$\vec{j} = Nq\dot{\vec{r}}$$

$$\dot{x} = \frac{iq}{m\omega} E_{0x}\, e^{-i\omega t} + \frac{q\Omega_c}{m\omega^2} E_{0y}\, e^{-i\omega t} \qquad \frac{iq\Omega_c}{m\omega^3}(-i\omega)E_{0x}$$

$$\dot{y} = \frac{iq}{m\omega} E_{0y}\, e^{-i\omega t} - \frac{q\Omega_c}{m\omega^2} E_{0x}\, e^{-i\omega t}$$

$$j_x = \frac{iNq^2}{m\omega} E_{0x}\, e^{-i\omega t} + \frac{Nq^2\Omega_c}{m\omega^2} E_{0y}\, e^{-i\omega t}$$

$$j_y = -\frac{Nq^2\Omega_c}{m\omega^2} E_{0x}\, e^{-i\omega t} + \frac{iNq^2}{m\omega} E_{0y}\, e^{-i\omega t}$$

$$\vec{j} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix} \begin{pmatrix} E_{0x} \\ E_{0y} \end{pmatrix} e^{-i\omega t}$$

Conductivity tensor $\quad \sigma_{ij} = \frac{iNq^2}{m\omega}\delta_{ij} + \frac{Nq^2}{m}\frac{\Omega_c}{\omega^2}\varepsilon_{ijz}$

$$\Rightarrow \quad \varepsilon_{ij} = \varepsilon_0\,\delta_{ij} + \frac{i}{\omega}\sigma_{ij}$$

$$= \varepsilon_0\left(1 - \frac{\omega_p^2}{\omega^2}\right)\delta_{ij} + i\,\varepsilon_0\,\frac{\omega_p^2\,\Omega_c}{\omega^3}\,\varepsilon_{ijz}$$

Since $\varepsilon_{ij} \neq \varepsilon \delta_{ij}$, have to study anisotropic medium!

$$\begin{cases} \vec{\nabla} \times \vec{E}(r,t) = -\frac{\partial}{\partial t} \vec{B}(r,t) \\ \vec{\nabla} \times \vec{H}(r,t) = \frac{\partial}{\partial t} \vec{D}(r,t) \end{cases}$$

$\vec{B}(r,t) = \vec{B}(r) e^{-i\omega t}$ and similarly for other fields.

$$\Rightarrow \begin{cases} \vec{\nabla} \times \vec{E}(r) = i\omega \vec{B}(r) \\ \vec{\nabla} \times \vec{H}(r) = -i\omega \vec{D}(r) \end{cases}$$

$$\vec{\nabla} \times \left(\vec{\nabla} \times \vec{E}(r)\right) = i\omega \vec{\nabla} \times \vec{B}(r) = i\omega \mu(\omega) \vec{\nabla} \times \vec{H}(r)$$

$$\vec{B}(r) = \mu(\omega) \vec{H}(r)$$

$$\Rightarrow \vec{\nabla}\left(\vec{\nabla} \cdot \vec{E}(r)\right) - \nabla^2 \vec{E}(r) = i\omega \mu(\omega) \left(-i\omega \vec{D}(r)\right)$$

$$\nabla^2 \vec{E}(r) = -\omega^2 \mu(\omega) \vec{D}(r)$$

But here $D_i(r) = \varepsilon_{ij}(\omega) E_j(r)$

$$\Rightarrow \nabla^2 E_i(r) = -\omega^2 \mu(\omega) \varepsilon_{ij}(\omega) E_j(r)$$

In the problem here the $\vec{r}$-dependence is via z-dependence (x, y dependence is absent due to uniformity)

$$\Rightarrow \frac{d^2}{dz^2} E_1(z) = -\omega^2 \mu(\omega) \left(\varepsilon_{11}(\omega) E_1(z) + \varepsilon_{12}(\omega) E_2(z)\right)$$

So that $$\frac{d^2}{dz^2} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = -\frac{\omega^2}{c^2} \frac{\mu(\omega)}{\mu_0 \varepsilon_0} \begin{pmatrix} \varepsilon_{xx} & \varepsilon_{xy} \\ \varepsilon_{yx} & \varepsilon_{yy} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix}$$

$\Rightarrow$ EM waves in an anisotropic medium

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \quad \text{where } \vec{D} = \varepsilon \vec{E}, \ \varepsilon = \varepsilon_b + \frac{i\sigma}{\omega}$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\nabla^2 \vec{E} + \vec{\nabla}\underbrace{(\vec{\nabla} \cdot \vec{E})}_{\emptyset} = -\frac{\partial}{\partial t}(\vec{\nabla} \times \vec{B})$$

$$\vec{\nabla} \times \vec{B} = \mu \vec{\nabla} \times \vec{H} = \mu \frac{\partial \vec{D}}{\partial t}$$

$$-\nabla^2 \vec{E} = -\mu \frac{\partial^2 \vec{D}}{\partial t^2} = -\mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2} = (-\mu\varepsilon)(-\omega^2)\vec{E}(z)e^{-i\omega t}$$

For transverse wave $\vec{E}(z,t) = \vec{E}(z)\, e^{-i\omega t}$

$\hookrightarrow E_x(z)\hat{x} + E_y(z)\hat{y}$

$$\Rightarrow \frac{d^2}{dz^2}\begin{pmatrix} E_x \\ E_y \end{pmatrix} = -\frac{\omega^2}{c^2}\frac{\mu\varepsilon}{\mu_0\varepsilon_0}\begin{pmatrix} E_x \\ E_y \end{pmatrix} = -\frac{\omega^2}{c^2}\left\{\left(1 - \frac{\omega_p^2}{\omega^2}\right)\delta_{ij} + \frac{i\omega_p^2 \Omega_c}{\omega^3}\varepsilon_{ijz}\right\}\begin{pmatrix} E_x \\ E_y \end{pmatrix}$$

$$\cancel{\mu\varepsilon = n^2 \mu_0 \varepsilon_0 = \frac{n^2}{c^2}}$$

Therefore for "vector of state" $\psi = \begin{pmatrix} E_x(z) \\ E_y(z) \end{pmatrix}$

$$\frac{d^2\psi}{dz^2} = \hat{M}\psi, \quad M = \begin{pmatrix} M_{xx} & M_{xy} \\ M_{yx} & M_{yy} \end{pmatrix}, \quad M_{yx} = -M_{xy}$$

Want to find eigenvectors $\hat{M}\psi = \lambda\psi$
(eigenstate of matrix M)

$$M_{xx} = -\frac{\omega^2}{c^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right) = M_{yy}$$

$$M_{xy} = -\frac{\omega^2}{c^2} \cdot \frac{i\omega_p^2 \Omega_c}{\omega^3} = -M_{yx}$$

Therefore

$$\begin{cases} (M_{xx}-\lambda)\,E_x + M_{xy}\,E_y = 0 \\ M_{yx}\,E_x + (M_{yy}-\lambda)\,E_y = 0 \end{cases}$$

$$\det\begin{pmatrix} M_{xx}-\lambda & M_{xy} \\ M_{yx} & M_{yy}-\lambda \end{pmatrix} = 0 \Rightarrow \lambda^2 - \lambda\left(M_{xx}+M_{yy}\right) + \left(M_{xx}M_{yy} - M_{xy}\cdot M_{yx}\right) = 0$$

$$\lambda_\pm = \frac{M_{xx}+M_{yy}}{2} \pm \frac{1}{2}\sqrt{\left(M_{xx}+M_{yy}\right)^2 - 4\left(M_{xx}M_{yy} - M_{xy}M_{yx}\right)}$$

$$= \frac{M_{xx}+M_{yy}}{2} \pm \frac{1}{2}\sqrt{\left(M_{xx}-M_{yy}\right)^2 + 4M_{xy}M_{yx}}$$

$$(M_{xx}=M_{yy}) \quad = M_{xx} \pm \frac{1}{2}\cdot 2\sqrt{-M_{xy}^2} = M_{xx} \pm \sqrt{-M_{xy}^2} =$$

$$= -\frac{\omega^2}{c^2} \pm \sqrt{\left(\frac{\omega^2}{c^2}\right)^2 - i^2\left(\frac{\omega_p^2\Omega_c}{\omega^3}\right)^2}$$

$$= -\frac{\omega^2}{c^2} \pm \frac{\omega^2}{c^2}\frac{\omega_p^2\Omega_c}{\omega^3}$$

$$= -\frac{\omega^2}{c^2} \pm \frac{\omega_p^2\Omega_c}{c^2\omega}$$

$$\frac{d^2\psi_+}{dz^2} = \lambda_+\psi_+$$

$$\psi_+(z) = \psi_+(0)\,e^{\pm i\sqrt{-\lambda_+}\,z}$$

$$-\lambda_\pm = \frac{\omega^2}{c^2}\left(1 \mp \frac{\omega_p^2\Omega_c}{\omega^3}\right)$$

$$\Rightarrow \sqrt{-\lambda_\pm} \cong \frac{\omega}{c}\left(1 \mp \frac{\omega_p^2\Omega_c}{2\omega^3}\right)$$

So that $\sqrt{-\lambda_\pm}$ is just wavevector $K_\pm$.

$$-\lambda_\pm = \frac{\omega^2}{c^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right) \mp \frac{\omega^2}{c^2}\frac{\omega_p^2\Omega_c}{\omega^3}$$

$$K_\pm = \sqrt{-\lambda_\pm} = \frac{\omega}{c}\sqrt{n^2 \mp \frac{\omega_p^2\Omega_c}{\omega^3}} \approx \frac{\omega n}{c}\left(1 \mp \frac{\omega_p^2\Omega_c}{2\omega^3 n^2}\right)$$

$$n^2 = 1 - \frac{\omega_p^2}{\omega^2}$$

What are eigenvectors $\psi_\pm$ corresponding to $\lambda_\pm$?

$$M_{xx} - \lambda_\pm = \mp\sqrt{-M_{xy}^2}$$

$$\mp\sqrt{-M_{xy}^2}\, E_x^\pm + M_{xy}\, E_y^\pm = 0$$

$$\frac{E_x^\pm}{E_y^\pm} = \frac{\pm M_{xy}}{\sqrt{-M_{xy}^2}} = \pm(-i) = \mp i.$$

$$\Rightarrow \psi_+(z) = E_x^+(z)\begin{pmatrix}1\\ i\end{pmatrix}, \quad \psi_-(z) = E_x^-(z)\begin{pmatrix}1\\ -i\end{pmatrix}$$

$$\vec{E}^{(\pm)}(z,t) = \left(E_x(z)\,\hat{x} + \left(\pm i\, E_x(z)\right)\hat{y}\right)e^{-i\omega t}$$

$$= E_x(z)\left(\hat{x} \pm i\,\hat{y}\right)e^{-i\omega t}$$

$\Rightarrow$ the eigenvectors are EM waves of right/left circular polarization !!!

Decomposition of the EM wave into $(\pm)$ components:

at $z=0$ arbitrary $\vec{E}(0,t) = E_x \hat{x} + E_y \hat{y} \rightarrow \psi(z=0,t)$

$$\psi(0) = A\psi_+(0) + B\psi_-(0)$$

Then for waves propagating in $(+z)$ direction (choosing $k = +i\sqrt{-\lambda}$)

at <u>later</u> $z>0$ $\psi(z,t) = A\psi_+(z,t) + B\psi_-(z,t)$

let $\vec{E}(0) = E_0 \hat{x} = \frac{1}{2}\left(E^{(+)}(z=0)(\hat{x}+i\hat{y}) + E^{(-)}(z=0)(\hat{x}-i\hat{y})\right)$

with $E^+(0) = E^-(0) = E_0$

Then $\vec{E}(z) = \frac{1}{2}\left(E^+(z)(\hat{x}+i\hat{y}) + E^-(z)(\hat{x}-i\hat{y})\right)$

$$E^+(z) \downarrow E_0 e^{ik_+z} \qquad E^-(z) \downarrow E_0 e^{ik_-z}$$

$$k_\pm = \frac{\omega n}{c}\left(1 \mp \frac{\omega_p^2 \Omega_c}{2n^2\omega^3}\right) = k_0 \mp q_c$$

$$\vec{E}(z) = \frac{1}{2}E_0\left(e^{i(k_0-q_c)z}(\hat{x}+i\hat{y}) + e^{i(k_0+q_c)z}(\hat{x}-i\hat{y})\right)$$

$$= E_0 e^{ik_0z}\left(\cos q_c z\,\hat{x} + i\hat{y}(-i\sin q_c z)\right)$$

$$= E_0 e^{ik_0z}\underbrace{\left[\cos(q_c z)\,\hat{x} + \sin(q_c z)\,\hat{y}\right]}_{\text{describes rotating polarization}}$$

Wavelength $\lambda_c = \frac{2\pi}{q_c} = \frac{2\pi}{k_0}\frac{2n^2\omega^3}{\Omega_c\omega_p^2} \gg \lambda_0 = \frac{2\pi}{k_0}$

$$q_c = k_0\frac{\omega_p^2\Omega_c}{2n^2\omega^3}$$

This is Faraday rotation.

Helicons

EM wave in electron gas subject to external magnetic field B_0

$$\varepsilon_{ij}(\omega) = \varepsilon_0 \delta_{ij} + \frac{i}{\omega}\sigma_{ij}(\omega)$$

We found
$$\sigma^{\perp}_{ij}(\omega) = \frac{e^2 n}{m}\begin{pmatrix} -i\omega + \frac{1}{\tau} & -\Omega_c \\ \Omega_c & -i\omega + \frac{1}{\tau}\end{pmatrix}\frac{1}{\Omega_c^2 + (-i\omega + \frac{1}{\tau})^2}$$

$$\sigma_{zz}(\omega) = \frac{e^2 n}{m}\frac{1}{-i\omega + \frac{1}{\tau}} \quad ; \text{ also } \omega_p^2 = \frac{e^2 n}{\varepsilon_0 m}$$

$$\Rightarrow \frac{i\sigma_{zz}}{\omega} = \varepsilon_0 \frac{i\omega_p^2}{\omega(-i\omega + \frac{1}{\tau})}$$

$$\frac{i\sigma^{\perp}}{\omega} = \varepsilon_0 \omega_p^2 \begin{pmatrix} 1 + \frac{i}{\omega\tau} & -\frac{i\Omega_c}{\omega} \\ \frac{i\Omega_c}{\omega} & 1 + \frac{i}{\omega\tau}\end{pmatrix}\frac{1}{\Omega_c^2 + (-i\omega + \frac{1}{\tau})^2}$$

We take the limit $\frac{1}{\tau} \ll \omega \ll \Omega_c \ll \omega_p$

Then
$$\frac{1}{\Omega_c^2 + (-i\omega + \frac{1}{\tau})^2} \approx \frac{1}{\Omega_c^2 - \omega^2} \approx \frac{1}{\Omega_c^2}$$

$$\frac{1}{\omega\tau} \ll 1.$$

$$\Rightarrow \varepsilon^{\perp}(\omega) \cong \cancel{\varepsilon_0 \delta_{ij}} + \varepsilon_0 \begin{pmatrix} \frac{\omega_p^2}{\Omega_c^2} & \frac{-i\omega_p^2}{\Omega_c \omega} \\ \frac{i\omega_p^2}{\Omega_c \omega} & \frac{\omega_p^2}{\Omega_c^2}\end{pmatrix}$$

and we can neglect $\varepsilon_0 \delta_{ij}$ in comparison with $\left(\frac{\omega_p}{\Omega_c}\right)^2 \gg 1$.

Also, $\varepsilon^{\parallel}(\omega) = \varepsilon_{zz}(\omega) = \varepsilon_0 \delta_{ij} + \varepsilon_0 \frac{i\omega_p^2}{\omega(-i\omega + \frac{1}{\tau})} \approx -\varepsilon_0 \frac{\omega_p^2}{\omega^2}$

Observe that $\frac{\omega_p^2}{\Omega_c \omega} = \left(\frac{\omega_p}{\Omega_c}\right)^2 \frac{\Omega_c}{\omega} \gg \left(\frac{\omega_p}{\Omega_c}\right)^2$

$$\Rightarrow \varepsilon_{xx} \ll \varepsilon_{xy} \ll \varepsilon_{zz}$$

$$\begin{cases} \vec{k}\times\vec{B} = -\mu_0\omega\vec{D} = -\dfrac{\omega}{\varepsilon_0 c^2}\vec{D} \\ \vec{k}\times\vec{E} = \omega\vec{B} \end{cases}$$

$$\Rightarrow \underbrace{\vec{k}\times(\vec{k}\times\vec{E})}_{\vec{k}(\vec{k}\cdot\vec{E})-\vec{E}k^2} = -\frac{\omega^2}{\varepsilon_0 c^2}\vec{D}$$

$$\Rightarrow \left(k^2\delta_{ij} - k_i k_j\right)E_j = \frac{\omega^2}{\varepsilon_0 c^2}\underbrace{\varepsilon_{ij}(\omega)E_j}_{D_i}$$

let us consider wave propagating along $\vec{B}_0 \Rightarrow \vec{k} = (0,0,k)$
$\hookrightarrow B_0\hat{z}$.

$$\Rightarrow (k^2 - k_z^2)E_z = 0 \Rightarrow D_z = 0 \Rightarrow E_z = 0.$$

The wave is transverse $\vec{E} = (E_x, E_y, 0)$, $\vec{k}\cdot\vec{E} = 0$.

$$k^2 E_x = \frac{\omega^2}{\varepsilon_0 c^2}\left(\varepsilon_{xx}E_x + \varepsilon_{xy}E_y\right) =$$

$$= \frac{\omega^2}{\varepsilon_0 c^2}\varepsilon_0\left(\frac{\omega_p^2}{\Omega_c^2}E_x + \frac{-i\omega_p^2}{\Omega_c\omega}E_y\right)$$

$$\begin{cases} \left(k^2 - \dfrac{\omega^2\omega_p^2}{c^2\Omega_c^2}\right)E_x + \dfrac{i\omega\omega_p^2}{c^2\Omega_c}E_y = 0 \\ \left(k^2 - \dfrac{\omega^2\omega_p^2}{c^2\Omega_c^2}\right)E_y - \dfrac{i\omega\omega_p^2}{c^2\Omega_c}E_x = 0 \end{cases}$$

$$\begin{bmatrix} & \\ & \end{bmatrix}\begin{pmatrix} E_x \\ E_y \end{pmatrix} = 0 \Rightarrow \det\begin{bmatrix} & \\ & \end{bmatrix} = 0$$

$$\Rightarrow \left(k^2 - \frac{\omega^2 \omega_p^2}{c^2 \Omega_c^2}\right)^2 - \left(\frac{\omega \omega_p^2}{c^2 \Omega_c}\right)^2 = 0$$

$$k^2 - \frac{\omega^2 \omega_p^2}{c^2 \Omega_c^2} = \pm \frac{\omega \omega_p^2}{c^2 |\Omega_c|}$$

But $\frac{\omega_p^2}{c^2}\left(\frac{\omega}{\Omega_c}\right)^2 \ll \frac{\omega_p^2}{c^2}\frac{\omega}{\Omega_c}$

$\Rightarrow$ can neglect $\left(\frac{\omega \omega_p}{c \Omega_c}\right)^2$!

$$\Rightarrow \omega = c^2 k^2 \frac{\Omega_c}{\omega_p^2}$$ (only $\oplus$ sign is physical for $\Omega_c > 0$)

1) The obtained dispersion is quadratic, $\omega \sim k^2$.
And the frequency is quite low since $\frac{\Omega_c}{\omega_p} \ll 1$.

2)
$$\begin{pmatrix} k^2 & \frac{i\omega \omega_p^2}{c^2 \Omega_c} \\ -\frac{i\omega \omega_p^2}{c^2 \Omega_c} & k^2 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = 0$$

$$\Rightarrow E_x = -\frac{i\omega \omega_p^2}{c^2 k^2 \Omega_c} E_y = -i E_y \quad \text{for } \Omega_c > 0$$

(and $E_x = i E_y$ for $\Omega_c < 0$)

$\Rightarrow$ the propagating EM wave is circularly polarized!

Right-handed polarization for $\Omega_c > 0$ (electrons)

$\rightarrow$ helicon (in plasma physics $\rightarrow$ whistling atmospherics)

Can be thought of as an "extreme" version of Faraday rotation

Recall The usual (no magnetic field, $B_0 = 0$) limit:

$$\Omega_c = 0 \Rightarrow \sigma_{ij} = \frac{e^2 n}{m} \begin{pmatrix} -i\omega + \frac{1}{\tau} & 0 \\ 0 & -i\omega + \frac{1}{\tau} \end{pmatrix} \frac{1}{\left(-i\omega + \frac{1}{\tau}\right)^2} = \frac{e^2 n \tau}{m} \frac{1}{1 - i\omega\tau} \delta_{ij}$$

is diagonal!

$$\Rightarrow \varepsilon_{ij}(\omega) = \varepsilon_0 \delta_{ij} + \frac{i\sigma_{ij}(\omega)}{\omega} = \left(\varepsilon_0 + \frac{i\sigma_0}{\omega(1 - i\omega\tau)}\right)\delta_{ij}$$

$$\frac{\varepsilon(\omega)}{\varepsilon_0} = 1 + \frac{i\sigma_0}{\varepsilon_0 \omega} \frac{1}{-i\omega\tau\left(1 + \frac{i}{\omega\tau}\right)} = 1 - \frac{\sigma_0}{\varepsilon_0 \tau \omega^2} \frac{1}{1 + \frac{i}{\omega\tau}}$$

$\Rightarrow$ For $\omega\tau \gg 1$ $\quad \frac{\varepsilon(\omega)}{\varepsilon_0} \simeq 1 - \frac{e^2 n}{\varepsilon_0 m} \frac{1}{\omega^2} = 1 - \frac{\omega_p^2}{\omega^2}$

so that $n^2(\omega) = \frac{\varepsilon(\omega)}{\varepsilon_0} = -\left(\frac{\omega_p^2}{\omega^2} - 1\right) < 0$ for $\omega < \omega_p$

$\Rightarrow$ the wave eqn $\quad k^2 \vec{E} = \frac{\omega^2}{\varepsilon_0 c^2} \varepsilon(\omega) \vec{E} = \frac{\omega^2 n^2(\omega)}{c^2} \vec{E}$

$$\Rightarrow k^2 = -\frac{\omega^2}{c^2}\left(\frac{\omega_p^2}{\omega^2} - 1\right) = \frac{-1}{c^2}\left(\omega_p^2 - \omega^2\right)$$

so that $k = i\varkappa$ is purely imaginary $\Rightarrow \varkappa = \frac{1}{c}\sqrt{\omega_p^2 - \omega^2}$ and describes transverse wave which decay exponentially fast into the bulk.

The helicon $\left(\Omega_c = \frac{eB_0}{m} > \omega,\ \Omega_c \ll \omega_p\right)$ describes the same medium subject to strong magnetic field B_0 and now the EM wave can propagate through the conductor!

Dielectric tensor and its properties

$$\varepsilon_{ij}(\omega) = \varepsilon_0 \delta_{ij} + \frac{i}{\omega} \sigma_{ij}(\omega)$$

Joule's heating $\quad P = \vec{j} \cdot \vec{E} \rightarrow j_i E_i$

$$\vec{j}(t) = \mathrm{Re}\left(\overleftrightarrow{\sigma}\, \vec{E}\right) \Rightarrow j_i(t) = \mathrm{Re}\left\{\sigma_{ij}(\omega) E_j e^{-i\omega t}\right\}$$

$$= \frac{1}{2}\left\{\sigma_{ij} E_j e^{-i\omega t} + \sigma_{ij}^* E_j^* e^{i\omega t}\right\}$$

$$\vec{E} \Rightarrow \mathrm{Re}(\vec{E}) = \frac{1}{2}\left\{E_i e^{-i\omega t} + E_i^* e^{i\omega t}\right\}$$

time averaged

$$\Rightarrow \overline{P} = \frac{1}{4}\left(\sigma_{ij} E_j E_i^* + \underbrace{\sigma_{ij}^* E_j^* E_i}_{\sigma_{ji}^* E_i^* E_j}\right) =$$

$$= \frac{1}{4}\left(\sigma_{ij} + \sigma_{ji}^*\right) E_j E_i^*$$

But $\quad \sigma_{ij} = -i\omega\left(\varepsilon_{ij} - \varepsilon_0\right)$

$$\sigma_{ij} + \sigma_{ji}^* = -i\omega\left(\varepsilon_{ij} - \varepsilon_0\right) + i\omega\left(\varepsilon_{ji}^* - \varepsilon_0\right) =$$

$$= -i\omega\left(\varepsilon_{ij} - \varepsilon_{ji}^*\right)$$

$$\overline{P} = \frac{-i\omega}{4}\left(\varepsilon_{ij} - \varepsilon_{ji}^*\right) E_i^* E_j$$

Now we parameterize

$$\varepsilon_{ij}(\omega) = \frac{1}{2}\left(\varepsilon_{ij} + \varepsilon_{ji}^*\right) + i\,\frac{1}{2i}\left(\varepsilon_{ij} - \varepsilon_{ji}^*\right)$$

$$= a_{ij}(\omega) + i\, b_{ij}(\omega).$$

Consider $a_{ji}^* = \frac{1}{2}\left(\varepsilon_{ji}^* + \varepsilon_{ij}\right) = a_{ij}$

$$b_{ji}^* = \frac{1}{-2i}\left(\varepsilon_{ji}^* - \varepsilon_{ij}\right) = \frac{1}{2i}\left(\varepsilon_{ij} - \varepsilon_{ji}^*\right) = b_{ij}$$

So that $\varepsilon_{ji}^* = a_{ji}^* - i\, b_{ji}^* = a_{ij} - i\, b_{ij}$

(Note that $\varepsilon_{ij}^* = a_{ij}^* - i\, b_{ij}^* \neq \varepsilon_{ji}^*$)

Thus $\overline{P} = \frac{1}{2}\omega \frac{1}{2i}\left(\varepsilon_{ij} - \varepsilon_{ji}^*\right) E_i^* E_j = \frac{1}{2}\omega\, b_{ij}(\omega) E_i^* E_j$

Under time-reversal: $\left.\begin{array}{l} t \to -t \\ \omega \to -\omega \\ B \to -B \end{array}\right\}$ but $\overline{P}$ must remain positive

$$\Rightarrow \quad b_{ij}(\omega)\Big|_B = -b_{ij}(-\omega)\Big|_{-B}$$

Via Kramers-Kronig we find $a_{ij}(\omega)\Big|_B = a_{ij}(-\omega)\Big|_{-B}$

<u>Check</u>

$$\varepsilon_{ij}(\omega) = \varepsilon_0 \begin{pmatrix} 1 - \frac{\omega_p^2}{\omega^2} & i\,\frac{\omega_p^2 \Omega_c}{\omega^3} \\ -i\,\frac{\omega_p^2 \Omega_c}{\omega^3} & 1 - \frac{\omega_p^2}{\omega^2} \end{pmatrix}$$

Onsager relation

$$\boxed{\varepsilon_{ij}(\omega)\Big|_B = \varepsilon_{ji}^*(-\omega)\Big|_{-B} = \varepsilon_{ji}(\omega)\Big|_{-B}}$$

$$\varepsilon_{12}(\omega) = \frac{i\,\omega_p^2 \Omega_c}{\omega^3}$$

$$\varepsilon_{21}^*(\omega) = \left(-\frac{i\,\omega_p^2 \Omega_c}{\omega^3}\right)^* = \frac{i\,\omega_p^2 \Omega_c}{\omega^3}$$

$$\varepsilon_{21}^*(-\omega)\Big|_{-B} = \frac{i\,\omega_p^2(-\Omega_c)}{(-\omega)^3} = \frac{i\,\omega_p^2 \Omega_c}{\omega^3} = \varepsilon_{12}(\omega)$$

Kramers–Kronig

$$\varepsilon'(\Omega) - \varepsilon_0 = \frac{1}{\pi} P\int_{-\infty}^{\infty} d\omega \, \frac{\varepsilon''(\omega)}{\omega - \Omega}$$

$$a_{ij}(\Omega) - \varepsilon_0 \delta_{ij} = P\int \frac{d\omega}{\pi} \, \frac{b_{ij}(\omega)}{\omega - \Omega}$$

$$a_{ij}(-\Omega)\Big|_{-B} - \varepsilon_0 \delta_{ij} = P\int_{-\infty}^{\infty} \frac{d\omega}{\pi} \, \frac{b_{ij}(\omega)\big|_{-B}}{\omega + \Omega} = (\omega = -\omega') =$$

$$= P\int_{-\infty}^{\infty} \frac{d\omega'}{\pi} \, \frac{b_{ij}(-\omega')\big|_{-B}}{-\omega' + \Omega} =$$

$$= P\int \frac{d\omega}{\pi} \, \frac{b_{ij}(\omega)\big|_{B}}{\omega - \Omega} = a_{ij}(\Omega)\Big|_{B} - \varepsilon_0 \delta_{ij}$$

$\Rightarrow$ obtain that $a_{ij}(\omega)\big|_B = a_{ij}(-\omega)\big|_{-B}$

$$\Rightarrow \left[\begin{aligned} \varepsilon_{ij}(\omega)\big|_B &= a_{ij}(\omega)\big|_B + i\, b_{ij}(\omega)\big|_B = \\ &= a^*_{ji}(\omega)\big|_B + i\, b^*_{ji}(\omega)\big|_B && \{\text{hermicity of } a_{ij},\ b_{ij} \\ &= a^*_{ji}(-\omega)\big|_{-B} - i\, b^*_{ji}(-\omega)\big|_{-B} && \{\text{oddness } \hat{b},\ \text{evenness } \hat{a} \\ &= \varepsilon^*_{ji}(-\omega)\big|_{-B} = \varepsilon_{ji}(\omega)\big|_{-B} && \{\text{reality} \end{aligned}\right.$$

$\Downarrow$

$\varepsilon^*_{ij}(+\omega) = \varepsilon_{ij}(-\omega)$ from $\varepsilon_{ij}(\omega) = \int dt\, e^{i\omega t} \varepsilon_{ij}(t)$

Our Drude's form $\varepsilon_{ij}(\omega) = \varepsilon_0 n^2 \delta_{ij} + \varepsilon_0 i \dfrac{\omega_p^2 \Omega_c}{\omega^3} \varepsilon_{ijz}$

has this property as $\varepsilon_{jiz} = -\varepsilon_{ijz}$

and $\Omega_c\big|_{-B} = -\Omega_c\big|_B$.

Now let us check the combination:

$$\varepsilon_{ij} - \varepsilon_{ji}^{*} = 0 \text{ for } i=j$$

$$\varepsilon_{12} - \varepsilon_{21}^{*} = \frac{i\,\omega_p^2 \Omega_c}{\omega^3} - \left(\frac{-i\,\omega_p^2 \Omega_c}{\omega^3}\right)^{*} = 0$$

$$\varepsilon_{21} - \varepsilon_{12}^{*} = 0$$

$\Rightarrow$ there ~~are~~ is no joule's heating (energy loss)!

Why? $\Rightarrow$ we set $\tau = \infty$ from the very beginning!!!